# Device9 - SHAP Feature Importance Comparison (All Features)

**Models**

| | HH_L1_weight | HH_L3_magnitude | HH_L5_magnitude | HH_L1_magnitude | HH_L3_weight | HH_L5_weight | HH_L1_mean | HH_L5_mean | HH_L3_mean | HH_L1_radius | BigFeat_Gen_0 | BigFeat_Gen_4 | BigFeat_Gen_1 | BigFeat_Gen_3 | HH_L1_std | HH_L5_std | HH_L5_radius | HH_L3_std | HH_L3_radius | BigFeat_Gen_2 | BigFeat_Gen_5 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ExtraTrees Original | 0.0296 | 0.0250 | 0.0249 | 0.0236 | 0.0232 | 0.0193 | 0.0169 | 0.0192 | 0.0173 | 0.0111 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0108 | 0.0073 | 0.0064 | 0.0069 | 0.0059 | 0.0000 | 0.0000 |
| ExtraTrees BigFeat | 0.0211 | 0.0151 | 0.0184 | 0.0154 | 0.0129 | 0.0152 | 0.0134 | 0.0115 | 0.0109 | 0.0087 | 0.0198 | 0.0145 | 0.0109 | 0.0114 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0112 | 0.0000 |
| TPOT Original | 0.0344 | 0.0306 | 0.0226 | 0.0295 | 0.0248 | 0.0187 | 0.0205 | 0.0184 | 0.0194 | 0.0089 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0083 | 0.0086 | 0.0091 | 0.0084 | 0.0090 | 0.0000 | 0.0000 |
| TPOT BigFeat | 0.0169 | 0.0155 | 0.0148 | 0.0115 | 0.0117 | 0.0140 | 0.0101 | 0.0078 | 0.0079 | 0.0073 | 0.0123 | 0.0131 | 0.0106 | 0.0081 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0088 |

**Features (Sorted by Average Importance)**

SHAP Value: 0.030, 0.025, 0.020, 0.015, 0.010, 0.005, 0.000